



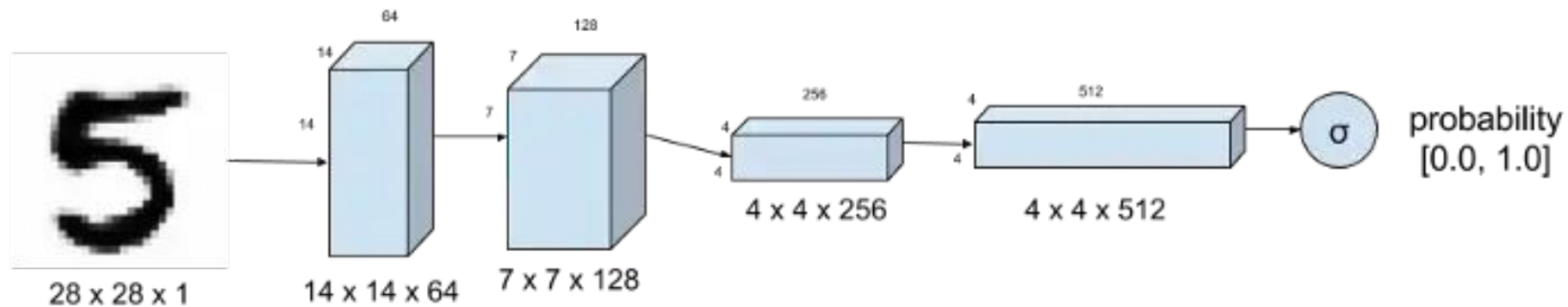
Generative Adversarial Networks (GANs)



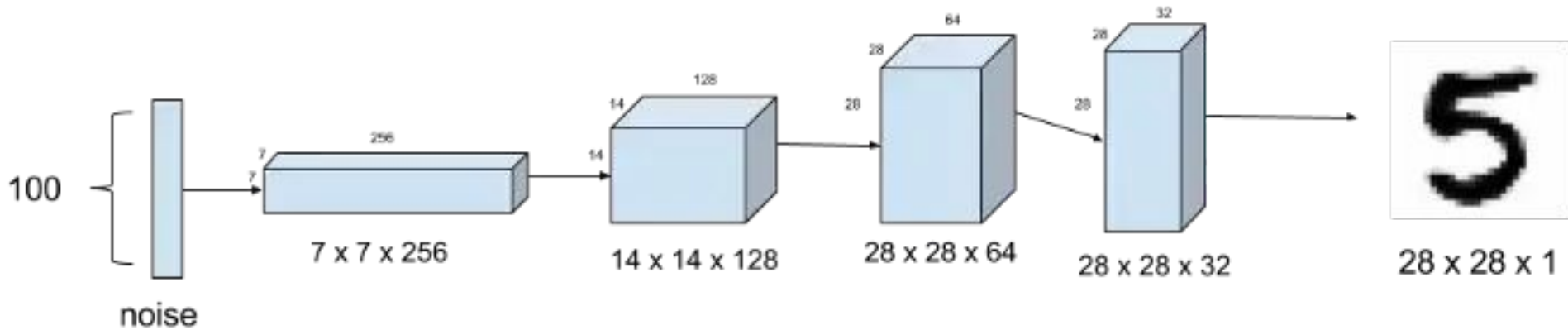
GANs

- 2 main components:
 - Discriminator
 - Generator

Discriminator



Generator



Conv and ConvTranspose

Input Kernel

0	1
2	3

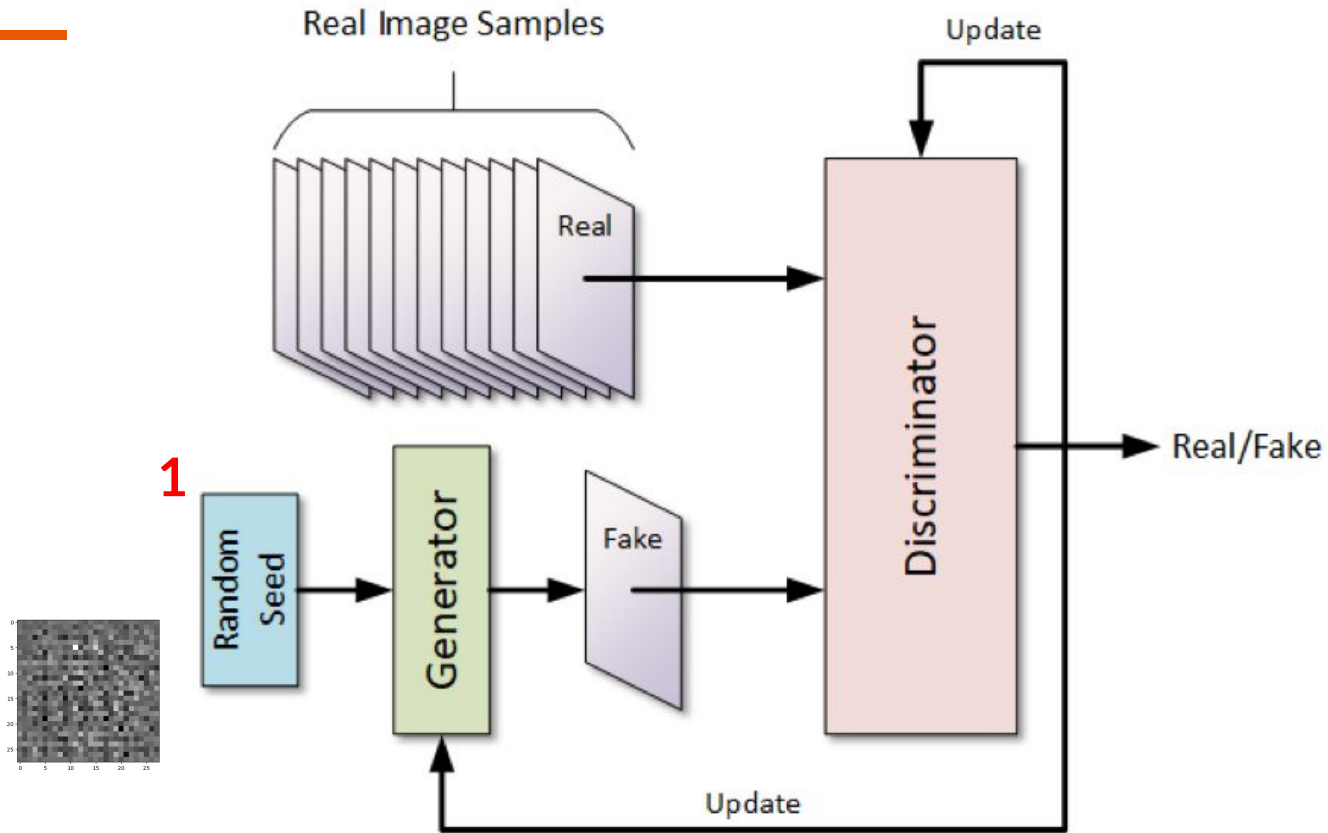
Transposed Conv (Stride 1)	
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4	1
2	3

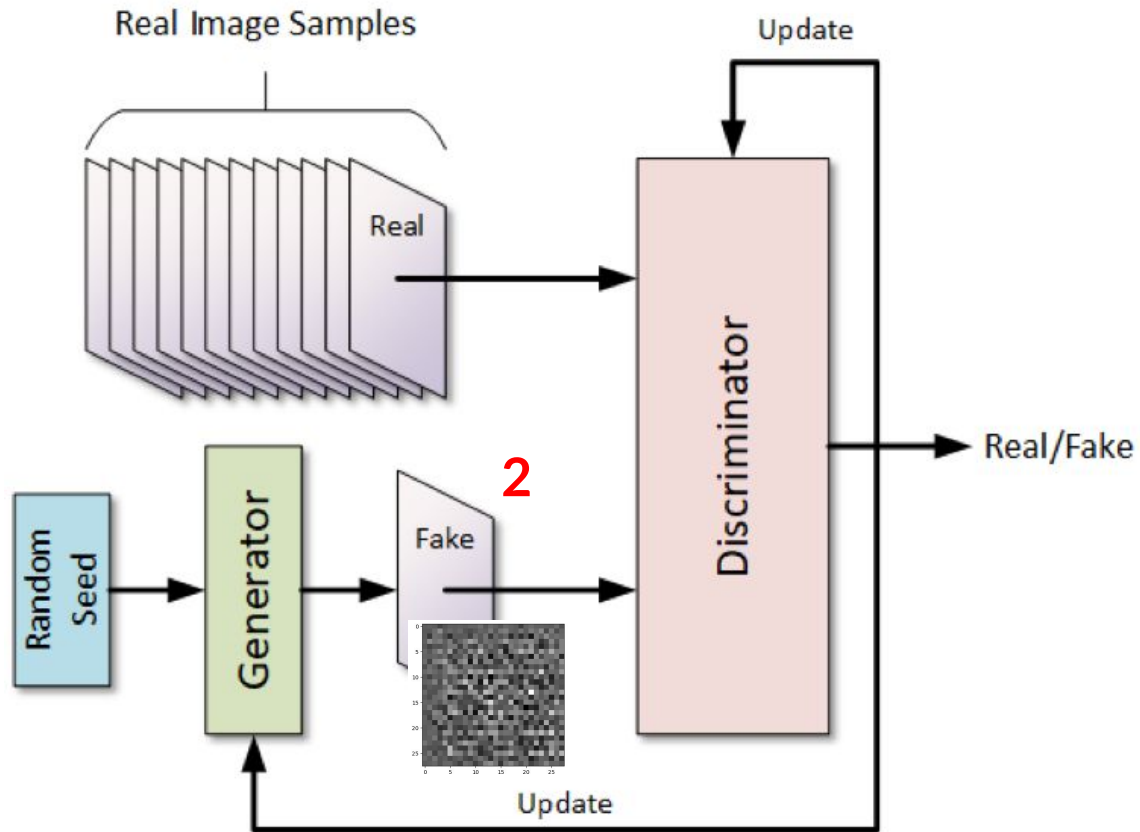
Output

$$= \begin{bmatrix} 0 & 0 & \\ 0 & 0 & \\ & & \end{bmatrix} + \begin{bmatrix} & 4 & 1 \\ & 2 & 3 \\ & & \end{bmatrix} + \begin{bmatrix} & & \\ 8 & 2 & \\ 4 & 6 & \end{bmatrix} + \begin{bmatrix} & & \\ & 12 & 3 \\ & 6 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 4 & 1 \\ 8 & 16 & 6 \\ 4 & 12 & 9 \end{bmatrix}$$

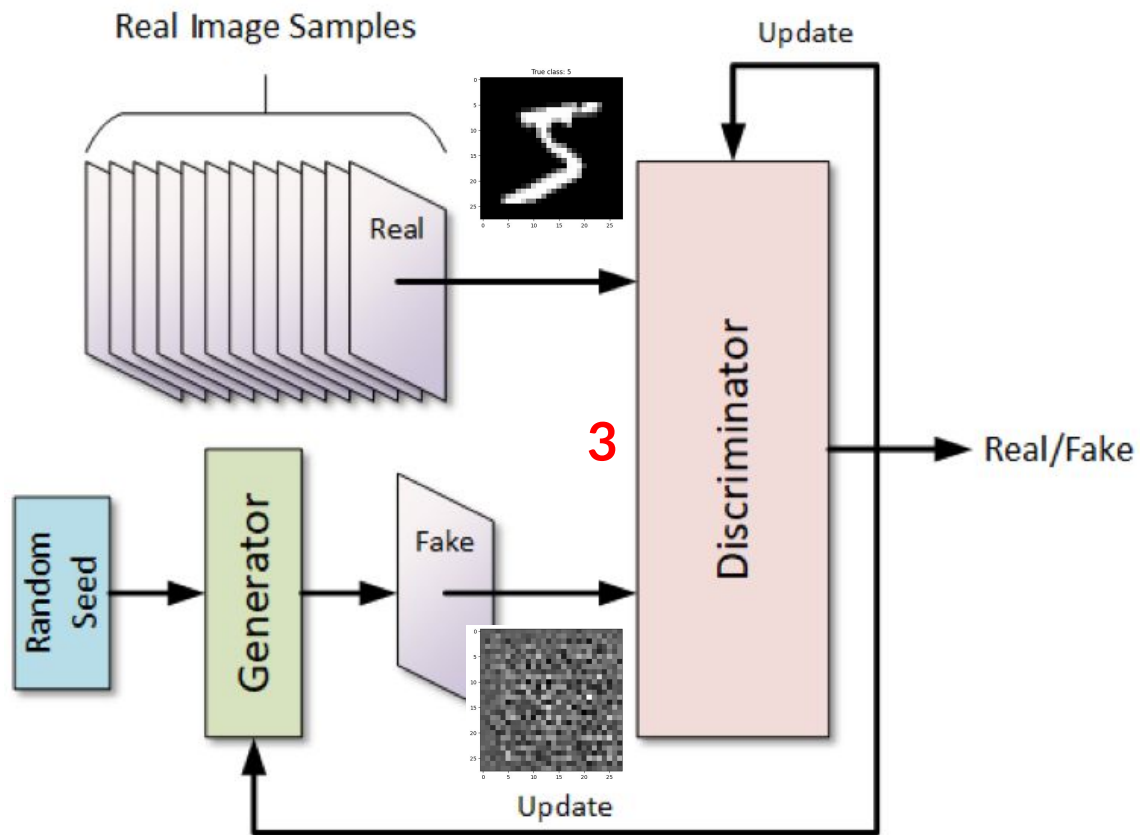
GAN



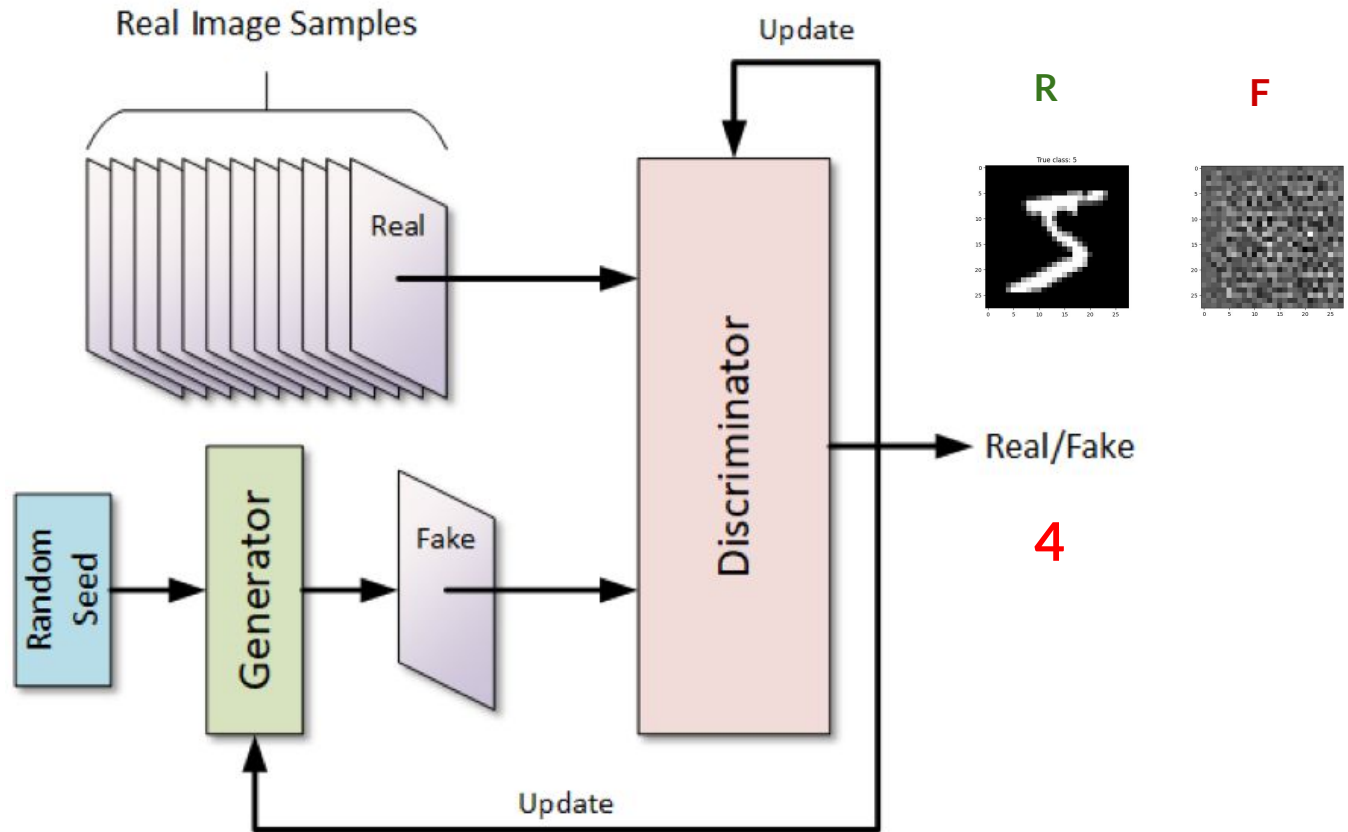
GAN



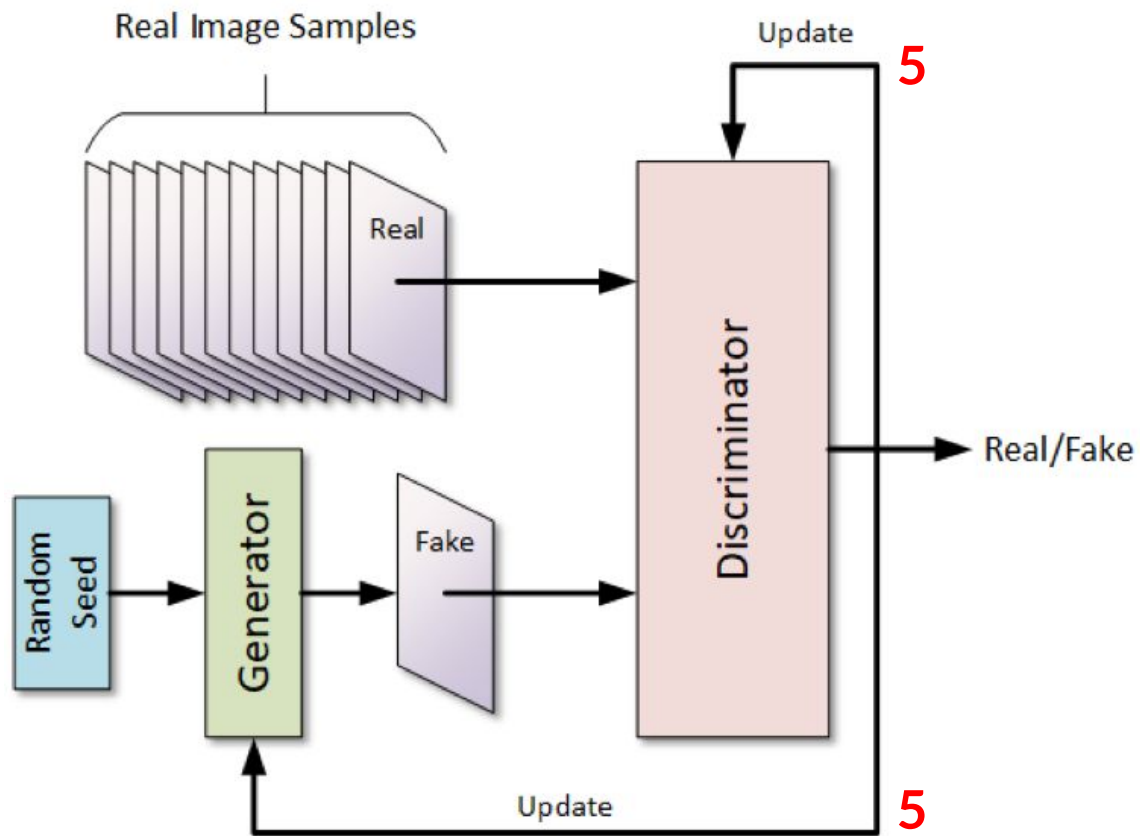
GAN



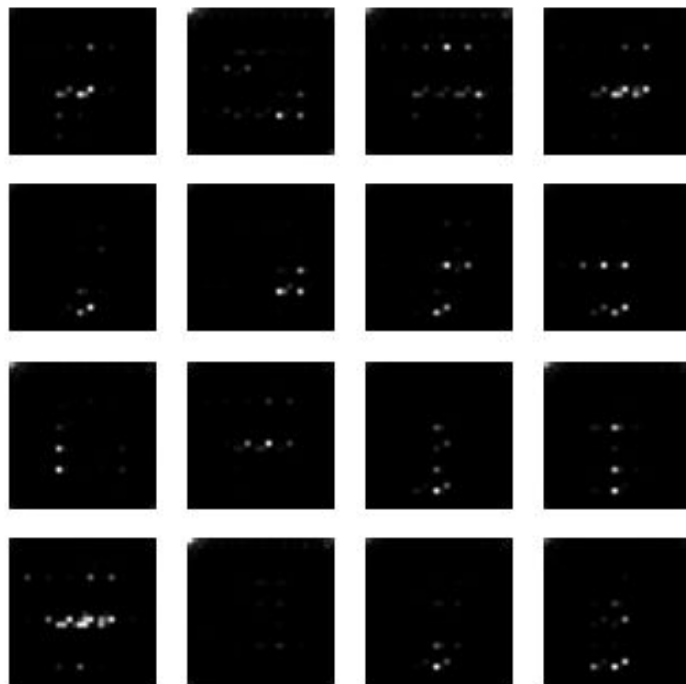
GAN



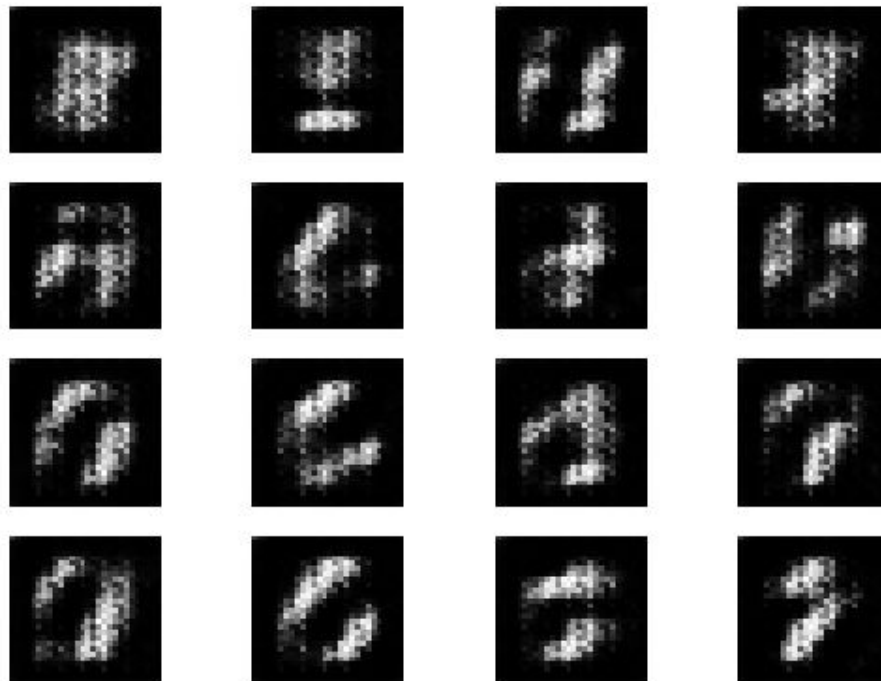
GAN



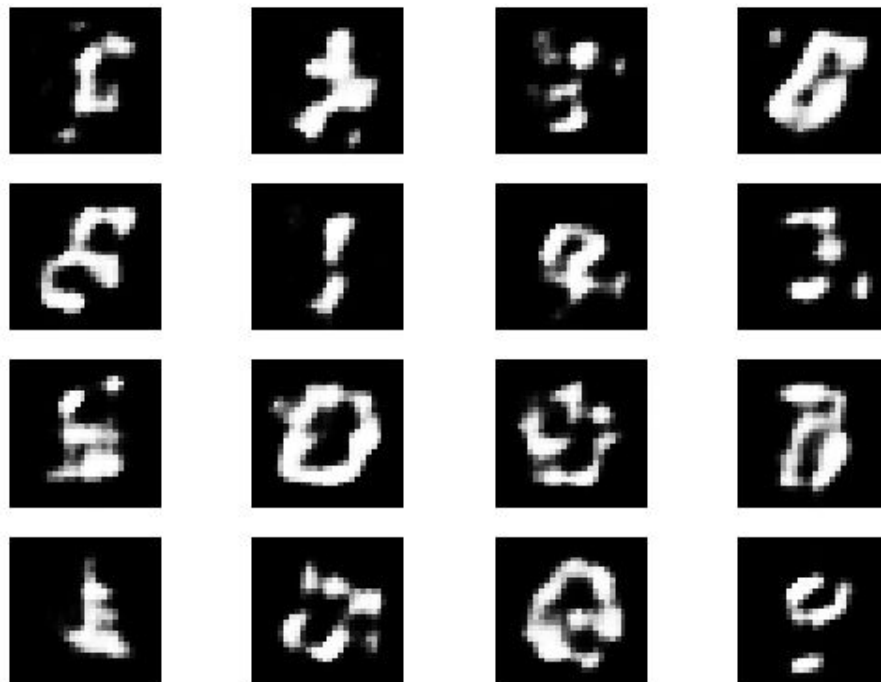
100 epochs



195 epochs



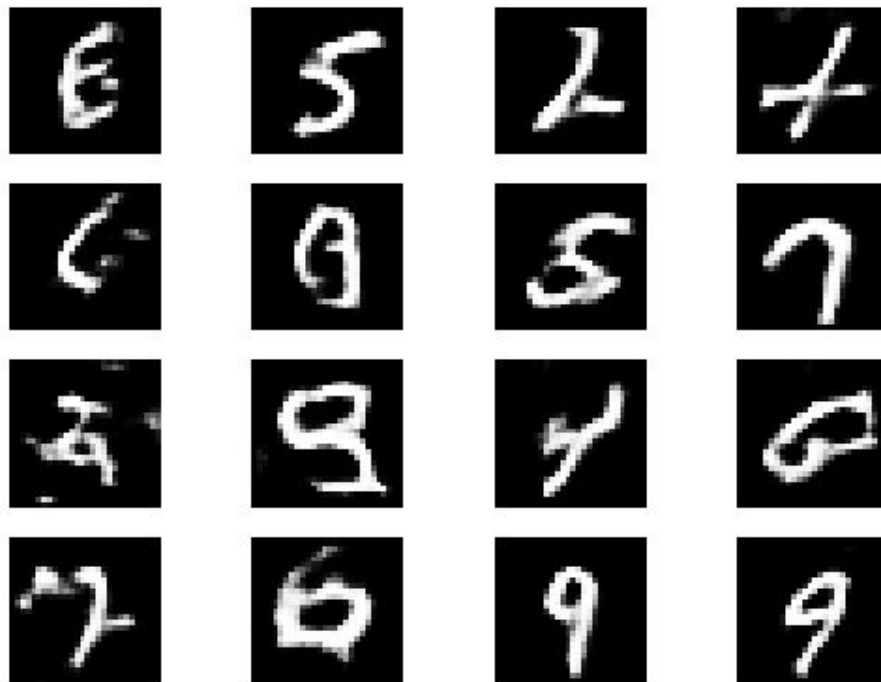
390 epochs



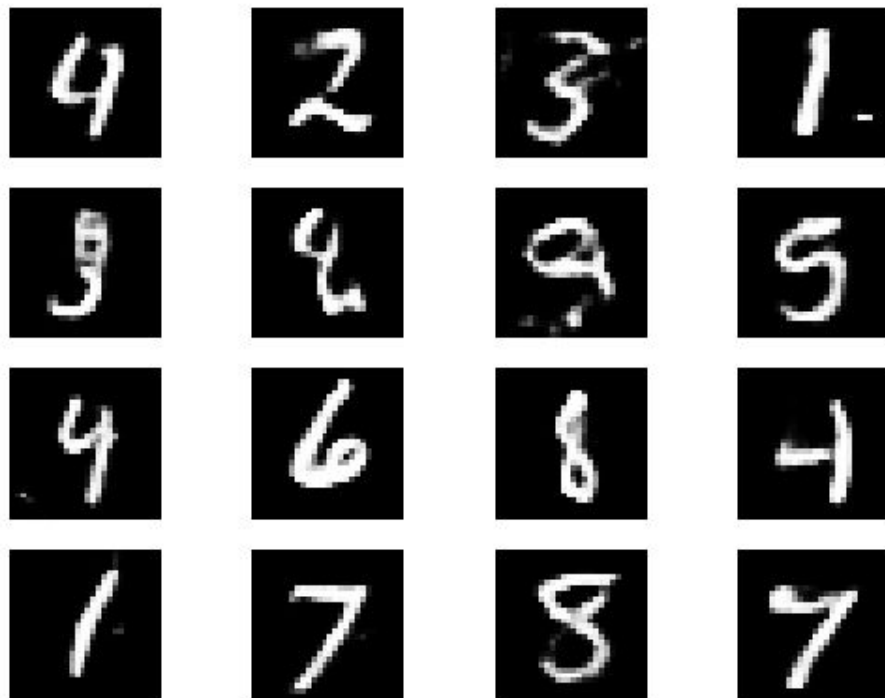
780 epochs



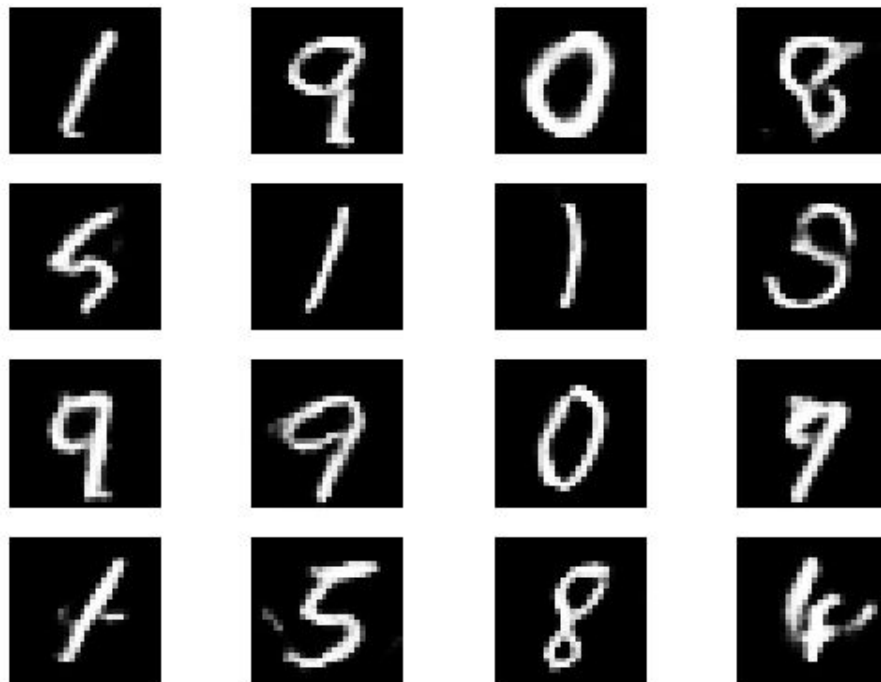
2145 epochs



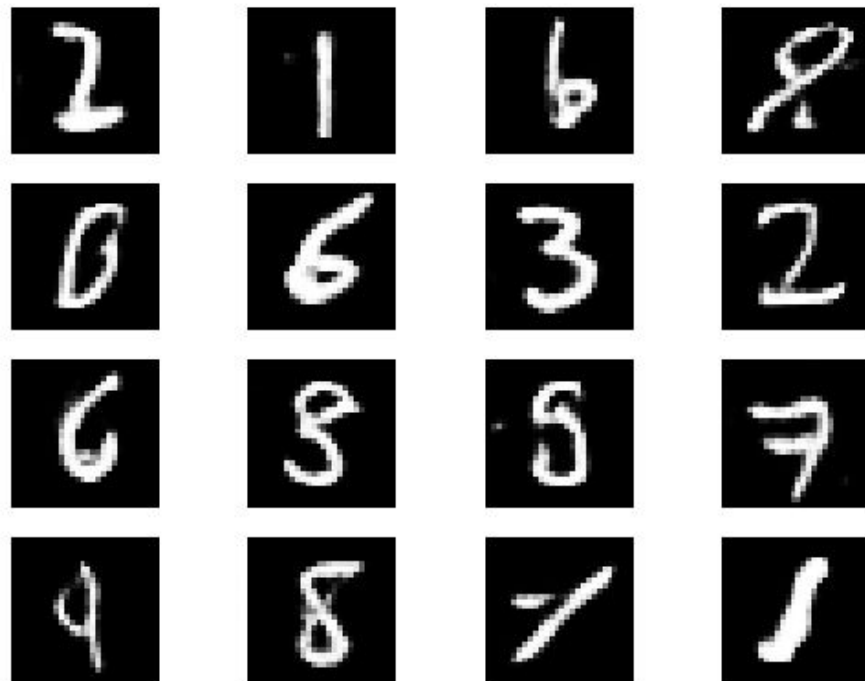
3900 epochs



7800 epochs



9750 epochs





Resources

- <https://www.tensorflow.org/tutorials/generative/dcgan>
- <https://machinelearningmastery.com/how-to-develop-a-generative-adversarial-network-for-an-mnist-handwritten-digits-from-scratch-in-keras/>
- <https://arxiv.org/pdf/1406.2661.pdf>